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Design and construction of MRT project Contract 825 of CCL1 in Singapore

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Contract 825 of CCL1 consists of 4 underground MRT stations and tunnels going under and adjacent to existing MRT lines and old buildings within difficult and changeable ground conditions. A summary of the scope and challenges that exist within the project is presented together with some of the key aspects of the design and construction of stations, which were used to overcome these challenges. Methods such as using top–down construction method with diaphragm walls as main soil retaining structure, extensive geotechnical instruments to monitor the deflection and settlement of diaphragm wall, adjacent ground and existing building and MRT structures. The instruments provide very useful information to control the site activity and reducing disturbance to existing structures. Underpinning work carried out at DBG station is also described in this paper.

All the four stations are built using the top–down construction method. A key consideration for choosing this construction method was to reduce soil movements and ground water lowering induced by the 28 to 35 m deep excavation, within close proximity to existing buildings and MRT structures. This together with requirements to reduce the disruption on the heavy traffic flow and disturbance to the environment led to the choice of top–down construction with diaphragm walls acting both as temporary and permanent soil retaining structure.

The selected method required a close integration between design and construction teams as the diaphragm walls were designed both for temporary and permanent conditions. The planning of the different traffic diversions to enable necessary access and openings for construction activities had also to be considered in the D-wall design. The design philosophy and principles using Plaxis analysis is described in Section 3.

The varying soil conditions called for different types of equipment for D-wall excavation. The experiences during D-wall installation are reflected in Section 4.

The extension of the Dhoby Ghaut station is carried out very close to the existing station and MRT tunnels imposing restrictions on the construction activities. Of special interest is the underpinning works for the B2-link between two existing stations on each side of the new one. This is described in Section 5.

As the project involves deep excavations and tunnels close to and under existing sensitive structures and MRT lines, instrumentation systems to provide feedback and assurance on design assumptions and predictions are important. The full range of geotechnical instruments has been employed on the contract with specific instrumentation schemes designed for specific structures. For the in-tunnel instrumentation, real time prism monitoring and real time track electrolevels have been utilized following the approach taken on earlier MRT projects, (Shirlaw et al 2000). For the historical surface structures individual schemes have been designed taking into account the predicted movement, the nature of the foundations, and settlement history, as far as feasible. These schemes involve limited but targeted real time electrolevels backed up by precise leveling and structure monitoring, together with recording of ground and diaphragm wall movements and piezometric changes within the various different strata. The monitoring is linked to sophisticated data collection allowing rapid processing into graphical format for ease of interpretation and dissemination to involved parties. This allows maximum benefit to be gained from the data in the form of controlling the construction works and feedback into the design process.

Keywords: Design and construction; Diaphragm wall; Instrumentation and monitoring; Underpinning; Jet grouting

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